

Generation Interconnection System Impact Study Report

For Queue Project Q15-0489

Pt. Mackenzie 100 kV

July 25, 2017

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q15-0489. The Interconnection Customer proposes a 139.4 MW solar photovoltaic facility interconnecting at the Pt. Mackenzie 100 kV substation in the Ohio area. The request was received on June 21, 2015. The study evaluated system intact and single contingency conditions using the regional planning case, including 183 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$2,354,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Jones Chapel-Bluebonnet 230kV	Robison Park-Russell Fork 230kV line	124.5	758	5.0
Pt. Mackenzie-Eastbridge 100kV L	Pt. Mackenzie-Sand Sta 100kV	108.4	103	100.0

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Bluebonnet-Jones Chapel 230kV	Rebuild line with higher-capacity conductor	\$30,855,000	2% (\$466,000)
Pt. Mackenzie-Eastbridge 100kV L	Reconductor line	\$1,888,000	100% (\$1,888,000)
	Total allocated to project		\$2,354,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.